

## AssignmentW7

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### Lab1-2

```
ashxio@ashio:~/public_html/naryvety.chey/lab07$ mkdir scripts data backups
ashxio@ashio:~/public_html/naryvety.chey/lab07$ touch data/file1.txt data/file2.txt data/file3.log
ashxio@ashio:~/public_html/naryvety.chey/lab07$ cd scripts/
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ nano print_args.sh
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ chmod +x print_args.sh
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ ./print_args.sh "one" "two three four" five
File name is "./print_args.sh"
1st arg is "one"
2nd arg is "two three four"
3rd arg is "five"
```

```
#!/bin/bash

echo "File name is \"$0\""
echo "1st arg is \"$1\""
echo "2nd arg is \"$2\""
echo "3rd arg is \"$3\""
```

### Lab3

```
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ nano print_all.sh
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ chmod +x print_all.sh
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ ./print_all.sh "one" "two 3 four" five six 7 8 9 10 11 a b c
File name is "./print_all.sh"
Total arguments: 12
Argument 1 is "one"
Argument 2 is "two 3 four"
Argument 3 is "five"
Argument 4 is "six"
Argument 5 is "7"
Argument 6 is "8"
Argument 7 is "9"
Argument 8 is "10"
Argument 9 is "11"
Argument 10 is "a"
Argument 11 is "b"
Argument 12 is "c"
```

```
#!/bin/bash

echo "File name is \"$0\""
echo "Total arguments: $#"
```

  

```
COUNT=1
for ARGS in "$@"
do
    echo "Arg $COUNT is \"$ARGS\""
    ((COUNT++))
done
```

## Lab4

```
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ nano print_dir.sh
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ chmod +x print_dir.sh
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ ./print_dir.sh "one" "../data/file1.txt" ../data ../backups/ 2
/
Dir 1 is "../data"
Dir 2 is "../backups/"
Dir 3 is "/"
```

```
#!/bin/bash

COUNT=1

for ARGS in "$@"
do
    if [ -d "$ARGS" ]
    then
        echo "Dir $COUNT is \"$ARGS\""
        ((COUNT++))
    fi
done
```

## Lab5

```
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ nano myscript.sh
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ chmod +x myscript.sh
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ ./myscript.sh "one" "../data/file1.txt" -a ../data ../backups/ 2
./ -b
Opt 1 is "-a"
Dir 1 is "../data"
Dir 2 is "../backups/"
Dir 3 is "/"
Opt 2 is "-b"
```

```
#!/bin/bash

OPT_COUNT=1
DIR_COUNT=1

for ARGS in "$@"
do
    if [[ -d "$ARGS" ]]
    then
        echo "Dir $DIR_COUNT is \"$ARGS\""
        ((DIR_COUNT++))
    elif [[ "$ARGS" == -* ]]
    then
        echo "Opt $OPT_COUNT is \"$ARGS\""
        ((OPT_COUNT++))
    fi
done
```

## Assignment W07

```
ashxio@ashio: ~/public_html/  x  +  v
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ nano backup_files.sh
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ chmod +x count_files.sh backup_files.sh
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ ./backup_files.sh ../data ../backups/
Backup completed successfully!
Files copied from '../data/' to '../backups/'
Number of files in '../data': 3
Number of files in '../backups/': 3
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ ./backup_files.sh -s ../data/ -b ../backups/
Backup completed successfully!
Files copied from '../data/' to '../backups/'
Number of files in '../data/': 3
Number of files in '../backups/': 3
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ ./backup_files.sh -b ../backups/ -s ../data/
Backup completed successfully!
Files copied from '../data/' to '../backups/'
Number of files in '../data/': 3
Number of files in '../backups/': 3
ashxio@ashio:~/public_html/naryvety.chey/lab07/scripts$ ./backup_files.sh ../data
Usage:
./backup_files.sh <source_dir> <backup_dir>
./backup_files.sh -s <source_dir> -b <backup_dir>
./backup_files.sh -b <backup_dir> -s <source_dir>
Examples:
././backup_files.sh ../data/ ../backups/
././backup_files.sh -s ../data/ -b ../backups/
././backup_files.sh -b ../backups/ -s ../data/
Notes:
- <source_dir> must exist and be a directory
- <backup_dir> will be created if it does not exist
- The script copies ALL regular files recursively and preserves subfolders.
```

```
localhost:8000/naryvety.chey/lab07/scripts/count_files.sh

#!/bin/bash

DIR="$1"
if [[ -z "$DIR" ]]; then
    echo "Usage: $0 <directory>"
    exit 1
fi
if [[ ! -d "$DIR" ]]; then
    echo "ERROR: '$DIR' is not a directory!"
    exit 1
fi
COUNT=0
for FILE in "$DIR"/*; do
    if [[ -f "$FILE" ]]; then
        ((COUNT++))
    fi
done
echo "Number of files in '$DIR': $COUNT"
```

```
localhost:8000/naryvety.chey/lab07/scripts/backup_files.sh

#!/bin/bash

usage() {
    echo "Usage:"
    echo " $0 <source_dir> <backup_dir>"
    echo " $0 -s <source_dir> -b <backup_dir>"
    echo " $0 -b <backup_dir> -s <source_dir>"
    echo "Examples:"
    echo " ./$0 ../data/ ../backups/"
    echo " ./$0 -s ../data/ -b ../backups/"
    echo " ./$0 -b ../backups/ -s ../data/"
    echo "Notes:"
    echo " - <source_dir> must exist and be a directory"
    echo " - <backup_dir> will be created if it does not exist"
    echo " - The script copies All regular files recursively and preserves subfolders."
    exit 1
}

if [[ $# -eq 2 ]]; then
    SOURCE="$1"
    BACKUP="$2"
elif [[ $# -eq 4 ]]; then
    if [[ $1 == "-s" && $3 == "-b" ]]; then
        SOURCE="$2"
        BACKUP="$4"
    elif [[ $1 == "-b" && $3 == "-s" ]]; then
        BACKUP="$2"
        SOURCE="$4"
    else
        usage
    fi
else
    usage
fi

if [[ ! -d "$SOURCE" ]]; then
    echo "ERROR: Source directory '$SOURCE' does not exist!"
    exit 1
fi

if [[ ! -d "$BACKUP" ]]; then
    mkdir -p "$BACKUP"
fi

cp -r "$SOURCE"/* "$BACKUP"

echo "Backup completed successfully!"
echo "Files copied from '$SOURCE' to '$BACKUP'"

./count_files.sh "$SOURCE"
./count_files.sh "$BACKUP"
```